



Exploring artificial intelligence and urban pollution emissions: "Speed bump" or "accelerator" for sustainable development?

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ABSTRACT

Within the framework of the overarching green development objective guiding the transformation and progress of urban green and low-carbon initiatives, the technological advancements stemming from artificial intelligence have introduced fresh paradigms for reducing urban pollution and enhancing environmental governance. A comprehensive exploration into the influence of artificial intelligence on urban pollution emissions has emerged as a pivotal current concern. The findings that the progression of artificial intelligence has effectively curbed urban pollution emissions, assuming the role of a significant "speed bump". Robustness testing corroborated this conclusion. Mechanism testing unveiled that the evolution of artificial intelligence mitigates urban pollution emissions by enhancing production efficiency, minimizing energy consumption, boosting green technology innovation, optimizing industrial structure, and increasing public participation. Heterogeneity analysis underscored substantial variances in the impact of artificial intelligence on urban pollution emissions, influenced by pollutant types, regional disparities, and urban scale differentials. Further scrutiny exposed that while the application of artificial intelligence technology does not entirely mitigate the "local-neighborhood" effect, it does ameliorate the "boundary effect", subsequently reducing pollution emission intensity in cities straddling provincial administrative boundaries.

1. Introduction

In the context of China's rapid ascent as the world's second-largest economy, its urbanization has brought about a myriad of social and economic benefits, accompanied by significant environmental challenges (Auffhammer and Carson, 2008; Fu et al., 2023). Urban pollution emissions have emerged as a formidable obstacle to national modernization drive, posing threats to public health (Daven, 2023), environmental stability (Hao, 2014), and overall economic progress (Chen et al., 2023). Data from China's National Energy Administration and the Bureau of Statistics show that in 2022, China's total energy consumption reached 5.41 billion tons of standard coal, marking a 2.9% increase from the previous year.¹ According to the BP Statistical Review of World Energy, China's carbon dioxide emissions escalated from 3.785 billion

tons in 2000 to 11.877 billion tons in 2022.² From this, it is evident that promoting green development characterized by ecological protection and energy conservation and emission reduction is urgent (Raihan, 2023; Bian et al., 2024) and cannot be delayed.

Artificial intelligence (AI), as an emerging technological innovation (Liu et al., 2022), offers a promising avenue for tackling urban environmental pollution, progressively evolving into a crucial tool to address the challenges posed by urban pollution emissions (Cowls et al., 2023). The report of the 20th National Congress of the Communist Party of China emphasized the need to "promote the high-end, intelligent, and green development of the manufacturing industry" and "promote the integrated cluster development of strategic emerging industries, constructing a batch of new growth engines in next-generation information technology, artificial intelligence, green environmental protection, and

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¹ The data is sourced from the "Statistical Bulletin of the People's Republic of China on the 2022 National Economic and Social Development".

² The data is sourced from the "BP Statistical Review of World Energy".

other areas". Significantly, China has experienced a notable surge in invention patents within the field of artificial intelligence, with applications and grants demonstrating substantial year-on-year growth. In 2022 alone, China's core artificial intelligence industry surged to an estimated 508 billion yuan, reflecting an 18% year-on-year increase as per the China Academy of Information and Communications³.

This rapid advancement in intelligence technologies offers new prospects for achieving smart cities, enhancing environmental efficiency (Germán et al., 2023), and mitigating pollution emissions. The rapid progress in intelligence technologies heralds a new era for smart cities, promising enhanced environmental efficiency and reduced pollution emissions. AI's widespread adoption is set to improve environmental monitoring accuracy, strengthen pollution source identification and control, optimize resource use, and further sustainable development goals (Lynn et al., 2022). While AI technology brings a "dividend effect" to environmental protection, its potential negative impacts on the environment should not be overlooked. Large-scale computing processes typically rely on data centers and servers, which consume significant amounts of electricity, directly leading to increased greenhouse gas emissions.

In this context, this study chooses to explore the potential and impact of AI technology in reducing urban pollution emissions from an urban perspective. Cities, being densely populated and industrially concentrated areas, are not only hotspots for pollution emissions but also the hubs where AI is most extensively utilized. Research by Qin and Wu (2015) indicates that carbon emission intensity shows a clear trend of initially increasing and then decreasing as the concentration of economic activities rises. Therefore, urban-level research can not only directly target the "disaster areas" of pollution but also contribute to more effective pollution control and management. Closely integrating AI technology with the spatial carriers of cities provides new perspectives and methods for addressing urban pollution emission problems, scientifically determining whether AI acts as a "speed bump" or "accelerator" in reducing urban pollution (Bacco et al., 2017).

2. Literature review

Artificial Intelligence has emerged as a transformative force across various sectors, profoundly impacting scientific research, industrial processes, and everyday life (Hoque and Khan, 2024). Its interdisciplinary nature spans machine learning, natural language processing, and computer vision, enabling innovations from automated decision-making to intelligent interaction systems (Nishant et al., 2020). Despite the rapid advancement of AI technologies offering numerous opportunities, it has also prompted significant research into its societal implications (Zhou et al., 2024), particularly regarding its effects on the economy, privacy, ethics, and employment. One area that has not received sufficient attention is AI's application in promoting environmental sustainability (Miller et al., 2023), especially in addressing urban pollution emissions. Although AI has demonstrated potential in environmental monitoring and pollution prediction, studies explicitly examining its impact on urban pollution are limited. This paper aims to explore AI technology's role in urban environmental management and its capacity to reduce pollution emissions, contributing new perspectives and methods for sustainable urban development.

The global focus on urban pollution emissions is intensifying, especially with the acceleration of urbanization, making the problem more severe (Francini et al., 2024). This issue is particularly pronounced in

developing and rapidly industrializing countries, where declining air quality and water pollution are the most significant environmental concerns (Reshmy et al., 2024). Current research primarily focuses on describing the spatial patterns and developmental trends of urban pollution emissions (Luo et al., 2017) and exploring how to reduce them through policies, technologies, and management measures (Li et al., 2022). These studies provide theoretical and empirical support for formulating effective environmental protection policies and implementing specific measures. However, the complexity and diversity of urban pollution issues necessitate more innovative and diverse research approaches, especially in the era of rapid information technology development.

With the advancement of technologies such as big data, artificial intelligence, and the Internet of Things, urban pollution monitoring and management are undergoing fundamental transformations (Zhang et al., 2021). The application of these technologies can not only improve the accuracy and real-time capabilities of pollution monitoring but also reveal the underlying causes of pollution through data analysis, supporting the development of more effective control strategies (Coccia and Bontempi, 2023). Supported by new technologies, urban pollution management is moving towards more intelligent and precise approaches. Intelligent monitoring systems can collect environmental data in real time and analyze it using cloud computing and AI to predict pollution trends, allowing for more dynamic and proactive pollution control (Roy, 2024). Additionally, enhancing public participation and awareness through smartphones, social media, and other channels offers new solutions for urban pollution management (Kumar et al., 2015; Andrei and Ioanid, 2023). Faced with the challenges of urban pollution emissions, a comprehensive application of interdisciplinary research methods and diverse technological tools is necessary. The effective combination of policy-making, technological application, and public participation can provide strong support for the sustainable development of urban environments. As new technologies continue to develop and be applied, future urban pollution management will become more efficient and intelligent, offering robust support for achieving environmental sustainability goals (Ye et al., 2020).

While numerous studies explore the intersection of AI and pollution on broader scales, such as national or industry levels, fewer delve into how AI affects urban pollution emissions within cities—the critical zones of AI development, application, and human survival. Based on data, investigating and addressing the impact of artificial intelligence on urban pollution emissions is a crucial foundation for reconciling the contradictions between economic growth and environmental protection. This study focuses on Chinese cities to examine AI's role in urban pollution, utilizing real-world data for theoretical and empirical analysis. It seeks to ascertain if AI serves as a "speed bump" or "accelerator" to urban pollution, thereby offering insights for technology markets and informing national strategies on environmental protection and sustainable growth.

3. Theoretical analysis and hypotheses

3.1. Artificial intelligence and urban pollution emissions

AI technology undoubtedly exerts a profound influence on urban pollution emissions. First, AI enables real-time monitoring of environmental parameters such as air quality, water quality, and noise through intelligent environmental monitoring systems (Selvaraj, et al., 2023). This facilitates a more precise understanding of pollution sources, pollutant distribution, and aids in formulating targeted policies by leveraging accurate data. Second, AI-driven transportation management optimizes traffic flow (Du et al., 2020), reduces congestion, and subsequently diminishes vehicle emissions. Intelligent traffic control systems and route planning, based on real-time data analysis, play a pivotal role in adjusting traffic patterns, thereby curbing fuel wastage and reducing exhaust emissions. Three, AI's application fosters green technology

³ The data is derived from the "AI Innovation Chain and Industrial Chain Integration Development Empowering the New Era of the Digital Economy—China Artificial Intelligence Patent Technology Analysis Report (2022)", published by the National Industrial Information Security Development Research Center and the Ministry of Industry and Information Technology's Electronic Intellectual Property Center

innovation, enabling the development of intelligent renewable energy systems and eco-friendly building designs, propelling cities towards greener and more sustainable trajectories.

However, the development of AI technology is not without its challenges. Notably, the energy consumption associated with AI is substantial, particularly during the training phase of deep learning models, necessitating a significant amount of computing resources and energy. The rapid evolution of AI technology leads to the obsolescence of electronic devices at a swift pace, generating a substantial amount of electronic waste, which can potentially elevate urban pollution emissions (Han and Chen, 2023). Additionally, in the era of big data, overreliance on AI for urban environmental monitoring and governance, especially when facing data deviations or technical issues, can result in significant resource waste and heightened energy consumption, consequently amplifying pollution emissions (Gao et al., 2020).

Based on these considerations, the following research hypothesis is proposed:

Hypothesis 1. Artificial intelligence technology has a significant inhibitory or promoting effect on urban pollution emissions.

3.2. Production efficiency mechanism

The impact of artificial intelligence on urban pollution emissions primarily manifests in enhancing labor productivity and optimizing capital efficiency. The widespread adoption of AI technologies can significantly boost productivity in production and service sectors by substituting laborious, monotonous, and hazardous tasks with automation, machine learning, and intelligent robotics (Damioli et al., 2021). This transition leads to a more efficient and precise production process, meaning more products or services can be generated in a given time frame while reducing the required resource input and corresponding pollution emissions (Liu et al., 2021). Moreover, AI can optimize capital efficiency by improving resource allocation, enhancing equipment utilization, and reducing production costs (Kamkar et al., 2024). Intelligent data analysis and forecasting enable businesses to better understand market demands and resource supply, avoiding surplus and shortage situations and maximizing the use of capital resources (Ababneh et al., 2023). Intelligent monitoring and scheduling reduce equipment downtime and increase the return on investment of capital. This optimization implies creating more value with less capital investment, thereby minimizing resource consumption and pollution emissions. In summary, the application of AI technologies enhances labor productivity and optimizes capital efficiency, leading to a relative reduction in resources and emissions required to produce a certain quantity of products or services. Such effects contribute to lowering urban pollution levels and promote the development of cities in a cleaner and more sustainable direction.

Hypothesis 2. The development of artificial intelligence reduces urban pollution emissions by improving labor productivity and optimizing capital productivity.

3.3. Energy consumption mechanism

AI technology has had a significant impact on urban pollution emissions, with energy consumption being a key aspect. AI can be applied to optimize and manage energy systems. By intelligently monitoring energy consumption and analyzing usage patterns, systems can develop more accurate energy utilization strategies (Garlik, 2017). This intelligent optimization can reduce unnecessary energy consumption, decreasing the demand for traditional, high-pollution energy sources and thereby lowering pollution emissions. Furthermore, AI technology can facilitate a shift in the structure of energy consumption. Intelligent data analysis and forecasting can enhance the understanding of the characteristics, supply and demand, and environmental impacts of different energy sources, thus promoting a transition to clean, renewable

energy. Intelligent technologies can help cities reduce their reliance on traditional, high-pollution energy sources like coal and oil, and increase the use of clean energy sources such as wind and solar power, reducing overall pollution emissions (Zhao et al., 2023). Additionally, AI can optimize the ways energy is consumed. Intelligent systems can achieve smart control and scheduling of energy, making its use more efficient and precise. For example, in the transportation sector, intelligent traffic management systems can optimize traffic flow, reduce congestion, and decrease vehicle idle times, thereby reducing fuel consumption and emissions. In summary, AI technology positively impacts urban pollution emissions through optimizing the structure of energy consumption, promoting a shift towards clean energy, and improving the ways energy is consumed. This impact helps drive cities towards cleaner, more sustainable energy utilization, making a significant contribution to improving environmental quality and reducing pollution.

Hypothesis 3. The development of artificial intelligence reduces urban pollution emissions by changing the traditional energy consumption structure and methods.

3.4. Green technology innovation mechanism

AI significantly impacts urban pollution emissions, particularly by facilitating green technology innovation (Yin et al., 2023). AI can accelerate the research and development of green technologies. Through intelligent data analysis, simulation, and forecasting, researchers can gain deeper insights into environmental issues, resource utilization, and climate change, providing accurate scientific foundations for innovating green technologies (Bijoria, 2024). AI optimizes the design and implementation of green technologies. Using AI algorithms and models, green technologies can be simulated and optimized to enhance their efficiency and feasibility. For example, in the renewable energy sector, AI can analyze weather and geographical data to optimize the layout of solar panels and wind turbines, maximizing the efficiency of energy collection and utilization while reducing pollution emissions (Ayenew and Berhanu, 2022). Additionally, AI technology promotes the intelligent application of green technologies. The development and deployment of concepts such as smart cities and intelligent transportation are supported by AI technology. Intelligent urban planning and management can better promote green modes of transportation, optimize traffic flow, and reduce congestion, thereby minimizing the impact of vehicle emissions on the urban environment. In summary, AI technology positively influences urban pollution emissions by promoting the innovation of green technologies, optimizing the design and implementation of these technologies, and facilitating their intelligent application. Therefore, the following hypothesis is proposed.

Hypothesis 4. The development of artificial intelligence reduces urban pollution emissions by promoting innovative development of green technology.

3.5. Industrial transformation and upgrading mechanism

The impact of artificial intelligence on urban pollution emissions is particularly significant in the context of industrial transformation and upgrading. First, AI technologies have facilitated the upgrade and optimization of industrial structures. Traditional industries, often reliant on high-energy consumption and high-emission production modes, are being reshaped by the widespread adoption of AI, promoting the development of high-tech, environmentally friendly industries. Emerging sectors such as intelligent manufacturing, new energy technologies, and energy conservation not only consume less energy but also reduce environmental pollution emissions, contributing to the improvement of urban environmental quality (Xu and Song, 2023). Second, AI enhances production efficiency, driving the transformation of traditional manufacturing towards intelligent, precise, and flexible production. This transformation not only reduces production costs and

improves product quality but also meets the diversified demands of the market. The proliferation of AI has given rise to new industries, such as big data, cloud computing, and intelligent manufacturing, providing new impetus for economic growth (Liu et al., 2022). Furthermore, AI promotes the optimization and collaboration of industrial chains, making supply chains more efficient and flexible, and enhancing the competitiveness of the entire industrial system. Industrial transformation and upgrading, shifting from low value-added, high-energy consumption, and high pollution to high value-added, low-energy consumption, and low pollution, can significantly reduce carbon emission levels (Liao et al., 2021).

Hypothesis 5. The development of artificial intelligence reduces urban pollution emissions by promoting the upgrading and optimization of industrial structure.

3.6. Public participation mechanism

First, with the development of artificial intelligence, communication and participation platforms based on AI are increasingly becoming effective tools for enhancing public awareness and involvement in sustainable environmental development (Goralski and Tan, 2020). These platforms, by offering real-time data analysis, personalized information delivery, and interactive participation mechanisms, significantly promote the public's initiative and efficiency in environmental protection activities, thereby contributing to the reduction of urban pollution emissions. AI technology can process and analyze vast amounts of environmental data, providing a more intuitive and in-depth presentation of environmental pollution. Second, AI platforms can deliver personalized information services based on users' interests and behavioral habits. This means that each user receives information and knowledge about environmental protection that is most relevant and interesting to them. This personalized approach not only improves the rate of information acceptance and participation but also stimulates users' environmental consciousness and action. Because the information is closely related to their personal lives, users are more likely to feel the impact of their actions on the environment (Kumar et al., 2019). Three, AI communication and participation platforms enhance public engagement through interactive features such as online surveys and gamified environmental challenges. These features not only make environmental activities more enjoyable and attractive but also allow participants to immediately see the impact of their actions, enhancing their sense of satisfaction and achievement. More importantly, through these means, the public can directly participate in specific environmental protection actions, such as energy conservation and emission reduction, waste sorting, etc., which have a direct positive impact on reducing urban pollution emissions. Therefore, the following research hypothesis is proposed.

Hypothesis 6. The development of artificial intelligence reduces urban pollution emissions by increasing public awareness and participation in environmental sustainability efforts through AI-based communication and engagement platforms.

4. Research design

4.1. Empirical model

To explore the impact of artificial intelligence on urban pollution emissions and determine whether it acts as a "speed bump" or "accelerator". In this case, we'll build a bidirectional fixed-effects model to assess the relationship between AI development and urban pollution emissions. The model can be expressed as follows:

$$\ln pol_{it} = \alpha_0 + \beta_1 \ln AI_{it} + \beta_2 \ln gdp_{it} + \beta_3 \ln hum_{it} + \beta_4 \ln gov_{it} + \beta_5 \ln size_{it} + \beta_6 \ln ind_{it} + \beta_7 \ln city_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

Where i represents the city, and t represents time. The dependent variable pol_{it} represents the pollution emission intensity of city i at time t , measured comprehensively through the city's industrial "three wastes" (wastewater, waste gas, and solid waste pollution), following the method of Liu and Lin (2019). The core explanatory variable AI_{it} represents the level of artificial intelligence development in city i during year t . The remaining variables are control variables that affect the pollution emission intensity of the city. α_0 is the constant term in the regression model, $\beta_1 \sim \beta_7$ represents the coefficient value of the parameters to be estimated, μ_i and ν_t represent time and individual fixed effects, respectively, and ε_{it} is the random error term.

To further explore the mechanisms through which artificial intelligence (AI) impacts urban pollution emissions, this study examines how AI influences urban pollution through production efficiency, energy consumption efficiency, green technology innovation, industrial transformation and upgrading, and public participation and oversight. Consequently, the following model is established to study the impact of AI on intermediary variables (M), namely production efficiency, energy consumption efficiency, green technology innovation capacity, industrial transformation and upgrading, and the level of societal participation, as detailed below:

$$\ln M_{it} = \sigma_0 + \delta_1 \ln AI_{it} + \delta_2 \ln gdp_{it} + \delta_3 \ln hum_{it} + \delta_4 \ln gov_{it} + \delta_5 \ln size_{it} + \delta_6 \ln ind_{it} + \delta_7 \ln city_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (2)$$

Given the potential correlation between production efficiency, energy consumption efficiency, green technology innovation capability, industrial transformation and upgrading, and the level of societal participation with urban pollution emissions, there may exist a high degree of collinearity between these factors. Confounding factors that could influence both the mediating variables and urban pollution emissions might lead to biased coefficient estimates. Although this stepwise regression method of testing mediation mechanisms is quite common, its problems have also been subject to scrutiny. Therefore, following Jiang's (2022) recommendations and practices regarding the testing of mediation mechanisms, this study will solely examine the relationship between artificial intelligence and the mediating variables.

4.2. Variable descriptions

4.2.1. Dependent variable

The dependent variable in this study is urban pollution emissions (pol). Based on the availability, accuracy, and continuity of city-level data, and following the approach of Wu et al. (2022), this study substitutes the emissions of industrial wastewater, industrial sulfur dioxide, and industrial solid waste pollution for wastewater, waste gas, and solid waste pollution, respectively. Sulfur dioxide, a major component of air pollution primarily originating from industrial production and energy consumption, can lead to the formation of acid rain, severely impacting the ecological environment and human health. Therefore, the emission of sulfur dioxide serves as a crucial indicator for assessing urban waste gas pollution and reflecting the quality of the urban atmospheric environment. Industrial solid waste pollution is proxied by the emission of industrial particulate matter, or dust and smoke, which directly relates to air quality. Emissions of particulate matter not only affect atmospheric visibility but can also cause respiratory diseases and other health issues, making it a comprehensive measure of urban atmospheric particulate pollution during industrial production.

Given that these three types of pollutants are not directly additive and have different units, the entropy method is further utilized to process the "three wastes" dimensionlessly, calculating weight coefficients to then aggregate into a comprehensive pollution emission index for each city each year. Urban pollution emissions are caused by various pollutants. Since this paper focuses on exploring the impact of artificial intelligence on urban pollution emissions, artificial intelligence, as a technological tool, directly influences the efficiency and method of

industrial production, thereby affecting the emission of industrial pollutants.

4.2.2. Independent variables

The core explanatory variable is artificial intelligence, which academia often measures using fixed asset investment in the computer software industry (Borland and Coelli, 2017), robot installation density (Liu et al., 2020), and patent bibliometric analysis methods (Hu, 2022). However, due to limitations in data acquisition, most studies are confined to the national, provincial, or industry levels. Analyzing the development level of artificial intelligence from the national or provincial perspective can be too broad, potentially leading to overestimation or underestimation of AI's impact on pollution emissions. Inspired by Yang's et al. (2023) approach to measuring the level of artificial intelligence at the provincial level, we use the interaction between the number of AI patent applications and the number of AI companies registered in each city within the same year to measure the level of artificial intelligence. This method more comprehensively reflects a city's development level and comprehensive strength in the field of AI. The interaction effect considers not only the capacity for technological research and innovation but also the actual situation of industrial development and application promotion, thereby more accurately measuring a city's AI level.

First, using the internationally recognized method of AI International Patent Classification (IPC), advanced searches are conducted through the China National Intellectual Property Administration using IPC classification codes (including: G10L, G06F, H04N, etc.) to obtain the number of AI patent applications (P) in province j . Second, for AI companies not categorized by patent type, this paper uses keywords such as artificial intelligence, machine learning, machine (computer) vision, computer recognition, virtual reality, neural networks, natural language processing, robotics, and biometrics to search for the number of newly registered AI companies (F) in city i in year t from 2005 to 2020 using the Qichacha and Tianyancha search platform. Finally, the level of artificial intelligence in city i at time t , AI , is represented as:

$$AI_{it} = P_{jt} \times \frac{F_{it}}{\sum_{j=1}^J F_{jt}} \quad (3)$$

In equation (3), $\frac{F_{it}}{\sum_{j=1}^J F_{jt}}$ represents the ratio of the number of artificial intelligence companies in city i to the total number of artificial intelligence companies across all cities within the same province as city i .

4.2.3. Mediating variables

Based on the research hypotheses, this paper involves the following mediating variables: production efficiency, energy consumption efficiency, green technology innovation capability, industrial transformation and upgrading level, and public participation.

①Production efficiency (includes two indicators). labor productivity (L) and capital productivity (K). Labor productivity is represented by the ratio of Gross Domestic Product (GDP) to the employed labor force, while capital productivity is calculated as the ratio of regional GDP to the regional fixed capital stock, representing the output created per unit of capital stock.

②Energy consumption efficiency (ec): Following the study by Wang (2023), energy consumption is measured in ten thousand tons of standard coal, with energy consumption per unit of GDP used to gauge energy efficiency. The conversion coefficients are 14.3 tec/10,000 m³ for natural gas, 1.7143 tec/ton for liquefied petroleum gas, and 1.229 tec/(10,000 kW h) for electricity.

③Green technology innovation capability ($inno$): Following the method of Chen and Chen (2021), this capability is represented by the logarithm of the number of city-level green patent grants, indicating the city's green technology innovation strength.

④Industrial transformation and upgrading (itu): In line with conventional methods, this is represented by the ratio of the total output of the tertiary industry to the total output of the secondary industry, indicating industrial structural upgrading.

⑤Public participation (pp): Considering the dependent variable of this paper is urban pollution emissions and examining the development of artificial intelligence through AI-based communication and participation platforms, whether it enhances public environmental participation awareness and supervision. With the rapid development of the internet, public life is inseparably linked to the internet, and internet search data recording netizens' user behavior can timely capture the public's attention to specific events. Therefore, following the method of Guo et al. (2020), the Baidu environmental pollution search index is used as a proxy for public environmental participation. "Environmental pollution" is used as the search keyword, and the daily average number of searches for "environmental pollution" obtained through the Baidu Index, with the annual average number of searches representing the level of public environmental participation.

4.2.4. Control variables

Based on previous literature, other variables that influence urban pollution emissions were chosen to be included. These variables are:

①Economic development level (gdp). Cities with higher levels of economic development often experience accelerated industrialization. The industrialization process involves extensive use of high-energy-consuming products such as fossil fuels, chemical industry products, and electricity. It's essential to control for the city's level of economic development when examining urban pollution emissions. This study measures the economic development level of cities using the logarithm of real per capita GDP, price-deflated to the base year of 2005.

②Education level (edu). The level of education affects citizens' environmental awareness, influencing social behavior norms and the implementation of environmental policies, thereby indirectly affecting urban pollution emissions. This study sums the average years of education for primary, junior high, high school, and college or higher education levels to calculate the average years of education per capita in each city.

③Government intervention (gov). The degree of government intervention reflects the government's effort in market regulation, which significantly impacts urban pollution emissions. This study uses the proportion of general budget expenditure to regional GDP to measure it.

④City size ($size$). As cities expand, the demand for population and resources grows, leading to increased energy consumption and industrial production scales. Therefore, this study uses the number of urban populations to measure city size.

⑤Industrialization level (ind). The level of urban industrialization is one of the key factors affecting urban pollution emissions, mainly because the production activities involved in industrialization often accompany pollutant emissions. This study characterizes it by the number of industrial enterprises in each city.

⑥Urbanization level ($city$). With the acceleration of urbanization, the urban population continues to increase, leading to a rapid rise in resource demand and energy consumption. It's necessary to control for the impact of urbanization level, measured in this study by the proportion of non-agricultural population to the total population.

To mitigate heteroscedasticity issues, all the above variables are log-transformed.

4.3. Data sources

This study focuses on 279 Chinese cities from 2005 to 2020, forming

4464 observational data points. Due to limitations in data availability, some city samples were excluded. Data for control variables affecting urban pollution emissions, "three wastes" (waste water, waste gas, and solid waste), and variables measuring production and energy consumption efficiencies are sourced from the annual China City Statistical Yearbook and the CEIC database (<https://insights.ceicdata.com.cn/>). Data on artificial intelligence patent applications are obtained from the official website of the China National Intellectual Property Administration (cnipa.gov.cn), and data on the number of green patent grants are sourced from the Chinese Research Database CNRDSH (<http://www.cnrds.com/>).

5. Empirical analysis

5.1. Benchmark regression

Table 1 provides the benchmark regression outcomes illustrating the impact of the artificial intelligence development level on urban pollution emission intensity. Columns (1) to (7) progressively introduce control variables while controlling for both city and year fixed effects. Remarkably, the estimated coefficients for $\ln AI$ remained steady within the range of 0.034–0.039 across these specifications, all displaying significant negative values. In Column (7) of Table 1, with the inclusion of a series of control variables, the estimated coefficient of $\ln AI$ remains significantly negative at the 1% statistical level. This suggests that for every 1% augmentation in the artificial intelligence development level within cities, pollution emission intensity decreases by 0.039%. The regression outcomes unequivocally assert that the advancement and application of artificial intelligence technology substantially curtail urban pollution emission intensity. As a result, Hypothesis 1 is substantiated, affirming that the evolution of artificial intelligence acts as a "deceleration belt" for urban pollution emissions rather than an "accelerator".

5.2. Verification of mechanisms

The mechanism of production efficiency suggests that the application of artificial intelligence technology can replace some manual operations, thereby enhancing work efficiency and capital utilization efficiency, and reducing urban environmental pollution. Consequently, this study constructs two indicators, labor productivity (L) and capital productivity (K), for separate regression analysis. The results, displayed

in columns (1) and (2) of Table 2, show the estimated outcomes of the regression analysis on the impact of AI development level on labor and capital productivity, respectively. The estimated coefficients of $\ln AI$ are significantly positive (0.0397 and 0.0648) and pass the 1% significance test, indicating that the development of AI has increased the city's labor and capital productivity. The advancement of AI can lead to higher labor and capital productivity, optimize resource use, improve transportation and logistics management, enable intelligent urban planning, and enhance energy efficiency, thereby reducing urban pollution emissions and promoting a more sustainable urban development path. Thus, Hypothesis 2 is confirmed.

The energy consumption efficiency mechanism examines the impact of the development level of artificial intelligence on energy consumption. The regression results reported in column (3) of Table 2 show that the estimated coefficient of $\ln AI$ is significantly negative (−0.095) and passes the significance test, indicating that the development of AI has reduced the consumption of traditional high-pollution, high-energy-consuming sources, thereby decreasing the intensity of urban pollution emissions. The advancement of AI can significantly reduce urban pollution emissions through various means, including improving energy utilization efficiency, promoting the application of clean energy technologies, optimizing the operation of energy systems, and encouraging energy-saving behaviors. Thus, Hypothesis 3 is confirmed.

The green technology innovation mechanism is analyzed in column (4) of Table 2, which presents the regression results of the impact of artificial intelligence development level on the level of green technology innovation. The findings indicate that AI significantly enhances the development level of green technology innovation in cities. The advancement of AI fosters innovation and application in green technologies, including clean energy, energy storage technologies, and low-emission transportation options. These green technologies can replace traditional high-emission energy sources and transportation methods, reducing environmental pollution. Thus, Hypothesis 4 is confirmed.

The mechanism of industrial transformation and upgrading is analyzed, with results shown in column (5) of Table 2. The estimated coefficient of $\ln AI$ is significantly positive (0.0489) and passes the 1% significance test, indicating that the application and development of artificial intelligence have promoted the transformation and upgrading of urban industrial structures. As a key representative of high-tech fields, AI's development has spurred the vigorous growth of related high-tech industries. Emerging industrial fields, such as intelligent manufacturing and green energy technologies, have benefited from the

Table 1
Benchmark regression results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable: $\ln pol$							
$\ln AI$	−0.094*** (−18.06)	−0.034** (−2.12)	−0.029* (−1.95)	−0.0265* (−1.80)	−0.034** (−2.28)	−0.038** (−2.51)	−0.039*** (−2.61)
$\ln gdp$		0.112*** (3.32)	0.310*** (11.44)	0.318*** (11.59)	0.289*** (9.72)	0.213*** (5.99)	0.218*** (6.14)
$\ln hum$			0.543*** (3.03)	0.538*** (3.00)	0.463** (2.55)	0.517*** (2.85)	0.505*** (2.79)
$\ln gov$				0.0407* (1.85)	0.0477** (2.15)	0.0431* (1.94)	0.041* (1.87)
$\ln size$					0.0787*** (2.66)	0.0755** (2.56)	0.068** (2.29)
$\ln ind$						0.0827*** (3.86)	0.076*** (3.52)
$\ln city$							0.075** (2.15)
Cons	2.269*** (152.95)	1.567*** (7.79)	−0.894** (−2.08)	−0.945** (−2.19)	−0.915** (−2.12)	−1.073** (−2.49)	−1.272*** (−2.90)
City	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	4464	4464	4464	4464	4464	4464	4464
R ²	0.225	0.494	0.478	0.478	0.478	0.480	0.484

Note: In the table, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, and the data in parentheses represent t-statistics.

Table 2
Mechanism inspection results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	lnL	lnK	lnec	lninno	lnitu	lnpp
lnAI	0.0397*** (2.81)	0.0648*** (2.71)	−0.095*** (−3.89)	0.329*** (16.78)	0.0489*** (4.83)	0.0692** (2.20)
lngdp	−0.348*** (−10.33)	0.331*** (5.99)	0.187*** (3.11)	0.104** (2.15)	−0.385*** (−15.40)	0.231*** (3.44)
lnhum	0.131 (0.76)	0.0908 (0.32)	1.274*** (4.10)	−1.009*** (−4.03)	0.666*** (5.17)	1.503*** (3.24)
lngov	−0.027 (−1.26)	−0.124*** (−3.54)	0.067* (1.86)	−0.165*** (−5.72)	−0.010 (−0.60)	−0.128*** (−2.93)
lnsize	0.0261 (0.92)	0.0253 (0.57)	0.557*** (9.99)	0.181*** (4.04)	0.0217 (0.94)	0.0502 (0.70)
lnind	0.149*** (7.25)	−0.421*** (−12.48)	−0.051 (−1.42)	0.189*** (6.59)	−0.205*** (−13.84)	0.0422 (0.97)
lncity	0.359*** (10.98)	−0.162*** (−2.90)	−0.011 (−0.20)	0.228*** (5.16)	−0.083*** (−3.61)	0.122 (1.20)
City	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Cons	0.606 (1.45)	3.200*** (4.80)	3.899*** (4.67)	0.896 (1.33)	3.102*** (8.94)	1.158 (0.93)
N	4464	4464	4464	4464	4464	4464
R ²	0.459	0.316	0.497	0.885	0.600	0.168

rapid development of AI technology, driving the industrial structure towards greener and more intelligent directions. The development of AI has facilitated the transformation and upgrading of the industrial structure, promoted green, intelligent, and clean technologies, optimized production and resource utilization efficiency, contributed to the construction of smart cities, and altered traditional industrial models, thereby significantly reducing urban pollution emissions. Thus, [Hypothesis 5](#) is confirmed.

The mechanism of public participation is explored through regression analysis, testing Model (2), with results presented in column (6) of [Table 2](#). The impact coefficient of artificial intelligence on public participation is 0.692 and passes the significance test at the 5% level, indicating that with the development of artificial intelligence, AI enhances public awareness and participation in environmental protection by providing convenient channels for information access. Additionally, AI can raise public awareness and skills in environmental protection through education and propaganda. Furthermore, AI can assist the public in environmental monitoring, promoting the implementation and execution of environmental protection policies. Therefore, AI, by increasing public participation and monitoring in environmental protection, can effectively reduce urban pollution emissions. [Hypothesis 6](#) is confirmed.

5.3. Heterogeneity analysis

5.3.1. Diverse forms of pollutants

Given the variation in different types of pollutants, a thorough examination of their heterogeneity in terms of pollution sources, transmission pathways, impact mechanisms, and governance methods is crucial. This comprehensive analysis is essential to understand how artificial intelligence may exert distinct effects on various forms of pollution. Such insights aid in rational resource allocation, prioritization of severe and impactful pollution types, and the minimization of damage to the urban environment.

To empirically evaluate these hypotheses, the study categorizes the explanatory variables into three pollutants: urban sulfur dioxide (SO_2), industrial wastewater (ww), and industrial smoke and dust (sm) based on their differing pollution characteristics, representing their emissions in logarithmic form. The specific regression outcomes are presented in columns (1) to (3) of [Table 3](#). Notably, the estimated coefficients of lnAI, −0.099, −0.288, and −0.354, are all statistically significant at the 1% level, reaffirming that the development of artificial intelligence indeed contributes to a reduction in urban pollution emissions. This reduction is

most pronounced in industrial smoke and dust pollution emissions, followed by industrial wastewater and urban sulfur dioxide. This pattern may be attributed to the intricate and diverse sources of industrial smoke and dust pollution, encompassing variables like particle size, shape, and composition. Artificial intelligence excels in processing and analyzing large-scale, high-dimensional complex data, effectively addressing this complexity. Furthermore, industrial smoke and dust emissions tend to be volatile, necessitating real-time monitoring and control. Artificial intelligence, leveraging technologies such as intelligent sensors and real-time data analysis, facilitates prompt monitoring and rapid response, ensuring emissions meet predefined standards.

5.3.2. Regional heterogeneity

Cities in different regions of China exhibit significant variations in natural resource endowments, geographical environments, development objectives, economic statuses, and government policy planning. These divergences may also influence the development level of artificial intelligence and its impact on the intensity of urban pollution emissions. Building upon this understanding, the study categorizes the urban sample into three subsets: eastern, central, and western, conducting regression estimations to observe how the development level of artificial intelligence affects urban pollution emission intensity in different regions.

The findings, displayed in columns (4) to (6) of [Table 3](#), reveal estimated lnAI coefficients of −0.079, −0.234, and −0.128 for urban samples in the eastern, central, and western regions, respectively. The development of artificial intelligence in cities across various regions acts as a "speed bump" on urban pollution emissions, playing a role in curbing pollution discharge. A comparative assessment of the coefficients highlights that as the artificial intelligence development level improves, the central region exhibits a more robust "deceleration effect" on urban pollution emission intensity, with the economically developed eastern region following closely. The relatively large and intricate industrial structure in the eastern region, along with a potentially slow transformation process, may hinder swift pollution reduction efforts. In contrast, the central region, commencing artificial intelligence technology implementation relatively later, can leverage and implement advanced, eco-friendly technological solutions, facilitating a more pronounced deceleration effect on pollution emissions through artificial intelligence technology promotion when compared to the eastern region.

Table 3
Heterogeneity test results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Heterogeneity of pollution forms			Heterogeneity of regional			Heterogeneity of urban scale		
	lnSO ₂	lnnw	lnsm	lnpol(Eastern China)	lnpol(Central China)	lnpol(Western China)	lnpol(Big -scale city)	lnpol(Medium -scale city)	lnpol(Small -scale city)
lnAI	−0.099*** (−3.92)	−0.288*** (−23.25)	−0.354*** (−18.59)	−0.079*** (−2.88)	−0.234*** (−10.81)	−0.128*** (−4.29)	−0.120* (−1.83)	−0.194*** (−4.12)	−0.023 (−1.30)
lngdp	0.092 (1.47)	0.302*** (9.40)	0.609*** (12.36)	0.361*** (5.19)	−0.050 (−0.86)	−0.001 (−0.01)	−0.280 (−1.59)	0.956*** (7.65)	0.100** (2.34)
lnhum	1.554*** (4.83)	−0.727*** (−2.62)	−0.0527 (−0.12)	0.464* (1.75)	0.553* (1.88)	3.009*** (6.05)	−2.121** (−2.21)	−1.643*** (−3.36)	0.313 (1.38)
lngov	0.120*** (3.24)	0.128*** (4.09)	−0.0672 (−1.40)	−0.182** (−2.35)	0.263*** (6.83)	0.0471 (0.97)	1.227** (2.24)	0.232 (0.81)	0.0276 (1.13)
lnsize	−0.103* (−1.78)	−0.044 (−0.88)	0.017 (0.22)	−0.004 (−0.10)	0.536*** (11.63)	−0.168** (−2.27)	−0.099 (−0.91)	−0.488*** (−3.66)	−0.013 (−0.34)
lnind	0.212*** (5.73)	0.0180 (0.63)	−0.143*** (−3.24)	0.175*** (4.46)	0.121*** (2.84)	−0.006 (−0.12)	0.286*** (2.73)	−0.0218 (−0.32)	0.047* (1.82)
lncity	0.169*** (2.97)	−0.190*** (−3.85)	−0.088 (−1.16)	0.160*** (2.59)	0.358*** (4.30)	−0.0774 (−1.21)	0.377 (1.55)	0.405*** (3.30)	0.012 (0.31)
Cons	5.062*** (5.85)	9.305*** (13.53)	7.622*** (7.22)	−2.822*** (−3.90)	−2.948*** (−4.23)	−3.857*** (−3.13)	7.518*** (2.72)	1.660 (1.22)	0.522 (0.88)
City	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	4464	4464	4464	1376	1264	1280	240	368	3856
R ²	0.691	0.257	0.106	0.579	0.229	0.411	0.746	0.596	0.461

5.3.3. Urban size heterogeneity

Cities of varying sizes may possess distinct primary sources of pollution and industrial structures. Larger cities often exhibit intricate and diverse industrial compositions, potentially hosting more high-emission industries. Conversely, smaller cities may predominantly engage in light industry or agriculture, resulting in a simpler emission structure. Consequently, when investigating the impact of artificial intelligence on urban pollution emissions, it is imperative to account for the heterogeneity in urban scale.⁴⁴ Subsample tests were performed based on samples from these different city sizes, and the specific results are illustrated in columns (7) to (9) of Table 3.

The outcomes reveal that the development level of artificial intelligence significantly diminishes urban pollution emissions in large- and medium-scale cities. Conversely, while the development of artificial intelligence also exhibits an inhibitory effect on pollution emissions in small-scale cities, this effect did not pass the significance test. Large- and medium-sized cities often have the capability to allocate more resources for the research, development, and application of artificial intelligence technologies, leading to relatively higher levels of AI development. This enables more precise pollution control and emission reduction. In contrast, small-scale cities, due to limited resources, may not be able to invest sufficiently in the development and application of AI technologies, affecting their ability to suppress pollution emissions. Furthermore, large- and medium-sized cities might have more complex and diverse industrial structures. High-pollution industries in these cities could be improved and optimized through AI technologies, thereby reducing emissions. Moreover, the industrial structure of small-scale cities may be simpler, with relatively fewer high-pollution industries, making the impact of AI technology on emissions less significant.

5.4. Further discussion

5.4.1. Cross-regional pollution transfer

When discussing the relationship between artificial intelligence and pollution emissions at the urban level, it's essential to delve into their

relationship from a spatial perspective. Urban pollution is not confined to local areas alone; pollutants can spread to surrounding regions or other cities through wind, water currents, and other means of diffusion. Previous research has highlighted phenomena such as cross-border pollution (Keller and Levinson, 2002; List et al., 2003), the migration of polluting industries across regions (Wu et al., 2017), pollution spillover to nearby areas, and the "pollution paradise" effect. Consequently, it's imperative to investigate whether artificial intelligence can aid regions in jointly addressing cross-regional pollution challenges in pollution control. This entails constructing the impact of artificial intelligence on "local neighboring" urban pollution emissions, utilizing a geographic distance spatial weight matrix⁵. Subsequently, the spatial Durbin model is utilized to estimate and examine the "local-neighborhood" effect between the development level of artificial intelligence and urban pollution emissions. The test results are presented in columns (1) and (2) of Table 4. The estimated coefficients of lnAI are −0.032 and 0.611, respectively, both passing the significance test. This indicates that with a 1% increase in the development level of local artificial intelligence, the intensity of local pollution emissions decreases by 0.032%, while pollution emissions in adjacent areas increase by 0.611%. It's evident that the phenomenon of pollution transfer persists, and the application and development of artificial intelligence technology have not fully mitigated the issue of pollution migration to nearby regions.

The transfer of pollution emissions to adjacent areas is a multifaceted challenge involving factors such as industrial structure, energy utilization, transportation, and urban planning. Solving this necessitates comprehensive efforts across various aspects and levels. A single technology is unlikely to provide a complete solution. Addressing pollution emission transfer to neighboring areas demands collaborative endeavors involving governments, enterprises, academia, and other stakeholders. This involves reinforcing research and technological applications, promoting broader integration of artificial intelligence in environmental protection, and enhancing regional cooperation and coordination to collectively address environmental governance challenges.

⁴ Utilizing the average urban population of 279 cities as the basis for categorization, cities are divided into three sizes: large-scale city (urban population >1 million), medium-scale city (urban population >500,000 and ≤ 1 million), and small-scale city (urban population ≤500,000).

⁵ The construction method for the geographic distance spatial weight matrix is as follows: the matrix's main diagonal elements are set to 0, and the non-main diagonal elements are the reciprocals of the geographic distances between cities *i* and *j*. Distance data is calculated using longitude and latitude coordinates between cities, applying spherical distance calculations

Table 4
Further discussion on inspection results.

Variables	(1) The native	(2) The neighborhood	(3) Provincial boundary cities	(4) Non-provincial border cities
lnAI	−0.032** (−2.30)	0.611*** (3.98)	−0.042** (−2.01)	−0.155*** (−8.23)
lngdp	−0.098*** (−2.92)	−1.450*** (−2.87)	0.113** (2.26)	0.085** (1.99)
lnhum	0.181 (0.93)	−5.903 (−0.67)	−0.263 (−1.03)	0.541** (2.53)
lngov	−0.003 (−0.13)	−0.457 (−0.36)	0.066** (2.34)	−0.104** (−2.47)
lnsize	−0.025 (−0.69)	−1.851 (−1.25)	−0.106** (−2.23)	0.240*** (8.16)
lnind	0.124*** (5.47)	2.109*** (4.85)	0.0555* (1.95)	0.250*** (10.33)
lncity	0.0382 (1.02)	−0.308 (−0.17)	0.077* (1.67)	0.028 (0.48)
rho	2.035*** (31.02)			
sigma2_e	0.101*** (47.16)			
Cons			2.012*** (2.94)	−2.212*** (−4.02)
N	4464		2800	1664
R ²	0.348		0.504	0.338

5.4.2. Boundary effect problem

Under a decentralized system, regional development exhibits boundary effects, where infrastructure and economic growth in areas at the borders of different administrative districts tend to lag behind due to the combined forces of environmental regulatory centrifugal forces, cost savings, and market demand centripetal forces. This leads to the clustering of polluting industries in border areas with weaker environmental regulations, thereby forming a pollution "grey edge" around central cities, known as the "boundary effect". Although the aforementioned test results confirm that the application of artificial intelligence in regions does not resolve the competitive relationship between local and adjacent city pollution emissions, the question remains whether the "boundary effect" observed in the middle zones between two places can be addressed. Further in-depth examination is required to answer this question.

To delve deeper into this issue, the entire sample is segregated into samples comprising cities adjacent to provincial boundaries and cities adjacent to non-provincial boundaries, facilitating a regression analysis. The test results, outlined in columns (3) and (4) of Table 4, indicate estimated coefficients of lnAI to be −0.042 and −0.155, respectively.

This indicates that the application and development of artificial intelligence technology can reduce urban pollution emissions, regardless of whether the city is located at provincial boundaries. The enhancement of AI development levels can help overcome the "boundary effect". AI has the capability to intelligently manage and optimize decisions. Whether a city is situated at the edge of a province or in other regions, intelligent systems can monitor environmental conditions and energy usage in real-time. Based on this data, these systems can make intelligent decisions to optimize resource allocation and energy use, reducing energy consumption and pollution emissions. The application and development of AI technology can provide intelligent, efficient solutions for cities across different regions, reducing pollution emissions, breaking through geographical boundaries, acting as a "speed bump" for pollution, and facilitating sustainable development.

5.5. Robustness and endogeneity discussion

5.5.1. Adopting winsorize

To mitigate the impact of extreme outliers, we employ a shrinkage treatment by adjusting the dependent variable, explanatory variable, and all control variables within a 1% deviation range, both upwards and downwards. We subsequently re-estimate the model based on equation

(1). The regression outcomes subsequent to this tail reduction treatment are presented in column (1) of Table 5. The table reveals a significantly negative estimation coefficient for lnAI at the 10% level of statistical significance, consistent with our benchmark regression results. This underscores that even after excluding extreme outliers, the findings presented in this article remain robust.

5.5.2. Delete municipality

Exclusion of central government-controlled municipality samples. Municipalities often possess distinctive attributes in terms of population, economy, and industrial structure that may not align with those of typical cities. This idiosyncrasy can potentially influence both data distribution and model outcomes. In the interest of data quality and comparability, we opt to exclude samples from directly governed municipalities such as Beijing, Tianjin, Shanghai, and Chongqing. Following this exclusion, we conduct a retest, the results of which are depicted in column (2) of Table 5. Remarkably, the regression results remain consistent with the aforementioned conclusions, confirming the robustness of our findings.

5.5.3. Endogeneity discussion

To mitigate estimation errors arising from potential endogeneity issues, we incorporate lags of one and two periods for the dependent variable and core explanatory variable. Additionally, we employ the instrumental variable method to re-estimate the model, thereby testing the robustness of the benchmark regression results. The regression outcomes are reported in column (3) of Table 5. The Arellano Bond test for AR (1) test statistic surpasses the 1% significance threshold, and the associated adjoint probability for the AR (2) test statistic exceeds 0.1. These results suggest the acceptance of the null hypothesis indicating no second-order autocorrelation and validate the reasonability of our endogeneity test results.

Furthermore, the significantly negative estimation coefficient for lnAI (−0.173) indicates that with a 1% increase in artificial intelligence development, urban pollution_{1q} emissions decrease by 0.173%, reaffirming the robustness of our conclusions. Leveraging the approach outlined by Han and Zhuang (2022), we utilize the density of industrial robots in the United States as an instrumental variable. Despite the United States' industry-level lead in robot applications compared to China, the two countries exhibit relatively similar development trends over the same period. Thus, we decompose the density of industrial robots in the United States into city-level data using the proportion of employment in each city's industry, estimating this variable as an instrumental variable. The results are detailed in columns (4) and (5) of Table 5. Both 2SLS estimation and GMM estimation consistently align with the benchmark regression, further affirming the robustness of our paper's conclusions.

6. Conclusion and implications

6.1. Conclusion

The rise, application, and development of artificial intelligence have profound impacts on the economy and society, while also presenting both opportunities and challenges for urban pollution emissions and their management. This study employs panel data from 279 Chinese cities spanning from 2005 to 2020 for empirical analysis, aiming to elucidate the impact of artificial intelligence development on urban pollution emissions and discern whether it acts as a "accelerator" or "speed bump".

The research reveals that the development level of artificial intelligence substantially diminishes the intensity of urban pollution emissions, acting as a "speed bump". To effectively identify the impact of artificial intelligence development on the intensity of urban pollution emissions, we also conducted robustness tests through methods such as trimming tails, excluding samples of municipalities directly under the

Table 5
Robustness test results.

Variables	(1)	(2)	(3)	(4)	(5)
	Winsorize	Delete municipality	Lagging for one and two period	Additional tool variable(2SLS)	Additional tool variable(GMM)
lnAI	-0.027* (-1.84)	-0.028* (-1.70)	-0.173*** (-10.17)	-0.076*** (-2.96)	-0.076*** (-2.96)
lngdp	0.168*** (4.70)	0.078* (1.93)	-1.253*** (-29.85)	-0.248*** (-5.68)	-0.248*** (-5.68)
lnhum	0.550*** (3.01)	0.328 (1.57)	0.708 (1.06)	-0.110 (-0.62)	-0.110 (-0.62)
lngov	0.001 (0.03)	0.0315 (1.32)	-0.446*** (-4.71)	-0.078*** (-3.01)	-0.078*** (-3.01)
lnsize	0.072** (2.39)	-0.0441 (-1.18)	0.226*** (2.96)	0.264*** (11.36)	0.264*** (11.36)
lnind	0.082*** (3.76)	0.061** (2.52)	0.768*** (19.41)	0.296*** (17.58)	0.296*** (17.58)
lncity	0.072* (1.96)	0.066* (1.80)	1.362*** (8.99)	-0.165*** (-3.73)	-0.165*** (-3.73)
Cons	-1.124** (-2.51)	0.552 (0.98)	-2.345* (-1.70)	1.476*** (2.91)	1.476*** (2.91)
AR (1)			-7.74 (0.000)		
AR (2)			-1.48 (0.139)		
Wald chi2(7)			6099.07	1396.14	1396.14
City	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	4464	4400	4464	4464	4464
R ²	0.480	0.480	0.362	0.215	0.215

central government, and using instrumental variable techniques, with results remaining significant. The mechanism tests reveal that AI primarily reduces urban pollution emission intensity by improving labor productivity, capital productivity, reducing the consumption of traditional high-pollution, high-energy sources, promoting green technology innovation, industrial structural transformation and upgrading, and enhancing public participation. Additionally, heterogeneity test results indicate that the emission reduction effect of improved AI development levels is stronger for industrial particulate matter pollution, cities in the eastern and central regions, and large- and medium-scale cities. Further discussions from a spatial perspective focus on the "local-neighborhood" effects between cities and "boundary effect". Research findings show that AI development has not resolved the competitive relationship between cities regarding pollution emissions, with evident pollution transfer still occurring. However, the application and development of AI technology help reduce pollution emissions in provincial boundary cities, mitigating the boundary effect.

In summary, enhancing the level of AI development can significantly reduce the intensity of pollution emissions in cities, leading to greener and sustainable urban development.

6.2. Implications

- (1) Strengthening top-level policy design and creating a favorable environment for the optimized development of artificial intelligence is crucial in the new development context. For economies worldwide to achieve high-quality and green development, the role of artificial intelligence as a digital technology must be fully leveraged. On one hand, governments should accelerate the formulation and implementation of policies guided by green and intelligent development, establishing incentive policies to encourage intelligent transportation, promote smart city planning, intelligent environmental monitoring and management, and foster clean energy and smart energy management. On the other hand, enterprises should adhere to policy guidance, set clear sustainable development goals, increase investment in green intelligent technologies, adopt environmentally friendly production and supply chain management, and promote clean energy and intelligent energy management. They should aim to

improve labor productivity, capital productivity, and reduce the consumption of traditional high-pollution, high-energy sources. By enhancing energy efficiency through artificial intelligence technology and collaborating with enterprises and research institutions on green technology, collective efforts can significantly advance urban green technology innovation and development.

- (2) Given the regional and urban scale heterogeneity, differentiated governance strategies should be adopted. Environmental policies tailored to the specific conditions of each region, city size, industrial structure, and economic level should be developed. For economically developed large cities, it is important to leverage their leading role in carbon emissions reduction. Artificial intelligence technology should be used to foster low-carbon, green, intelligent, and digital development in modern industrial chains. For areas with lower economic levels, artificial intelligence technology should be employed to green and modernize traditional, extensive, and high-energy-consuming industries, promoting a transition towards clean energy and green industries. Additionally, establishing a learning network between cities, through experience-sharing platforms and cooperative governance projects, can enhance collaboration between and among cities and regions, encouraging a collective effort to address cross-city or cross-regional pollution and governance issues.

6.3. Limitations and research lines

This study, set against the backdrop of rapid advancements in big data, the internet, and the Internet of Things, investigates the impact and internal mechanisms of artificial intelligence on urban pollution emissions, addressing whether AI acts as a "accelerator" or "speed bump" to urban pollution emissions. Given our limited theoretical foundation and the challenges in acquiring relevant AI data, this paper acknowledges several areas for improvement. In future research, there is hope to collect data at the district and county levels through big data analysis to refine the granularity of data analysis and enhance the precision of empirical analysis. Furthermore, urban pollution encompasses more than just the "three wastes", including a variety of complex pollutants. Future studies aim to conduct a more comprehensive analysis of a wider range of urban pollutants, providing a deeper understanding of the

environmental impacts associated with urban development and the role of AI in mitigating or exacerbating these effects.

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Qingfeng Luo: Writing – original draft, Conceptualization. **Pencheng Feng:** Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of the manuscript entitled.

Data availability

The authors do not have permission to share data.

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